ms of Machine Learning

core learning frameworks

core learning frameworks

Loss functions, gradient descent, risk minimization

Labelled data → mapping function f(x) ≈ y

SUPERVISED LEARNING: PREDICTION WITH GROUND TRUTH

Supervised Learning: Prediction with Ground Truth

Labelled data → mapping function f(x) ≈ y

Unsupervised Learning: Structure Without Labels

Clustering, dimensionality reduction

Discovering latent patterns in raw data

Clustering, dimensionality reduction

Reinforcement Learning: Learning from Interaction

Agent, environment, reward signal

Policy optimization, value functions

Agent, environment, reward signal

SUPERVISED USE CASES (BEYOND SPAM FILTERING)

Supervised Use Cases (Beyond Spam Filtering)

Semiconductor floorplanning with placement cost minimization

Diagnostic prediction from labeled medical imaging volumes

Application 1

Semiconductor floorplanning with placement cost minimization

Application 2

Diagnostic prediction from labeled medical imaging volumes

UNSUPERVISED USE CASES (BEYOND SPAM FILTERING)

Unsupervised Use Cases (Beyond Spam Filtering)

Molecular retrosynthesis pathway discovery via clustering

Seismic fault line detection from unlabeled geophone arrays

Application 1

Molecular retrosynthesis pathway discovery via clustering

Application 2

Seismic fault line detection from unlabeled geophone arrays

REINFORCEMENT LEARNING USE CASES (BEYOND SPAM FILTERING)

Reinforcement Learning Use Cases (Beyond Spam Filtering)

Automated portfolio rebalancing with Markov decision processes

Warehouse robot fleet coordination via multi-agent RL

Application 1

Automated portfolio rebalancing with Markov decision processes

Application 2

Warehouse robot fleet coordination via multi-agent RL

Workflow & Outlook

From raw data → curated features → model selection

Hybrid pipelines that stitch supervised ↷ unsupervised ↷ RL

From raw data → curated features → model selection